



Department of Computer Science

Assignment 1

Course Code: CS-201

Course Title: Data Structure

Total Marks: 15

Date:03-11-2025

Name: _____

Registration No. _____

Discipline/Semester: BSCS- 3rd A

Submission deadline: 07-11-2025

Instructions:

- Late submissions will not be entertained.
- Copied assignments will be marked ZERO.
- Use of AI tool is strictly not allowed.
- Submit the assignment in hard form must be handwritten and in A4 pages.

CLO1, GA2, C2

Question 1:

Interpret the following questions:

Part 1:

Consider the following multidimensional array:

X(1:5, 3:33)

Y(1:15, 10:20)

- Find the length of each dimension and the number of elements in X and Y.
- Suppose $\text{Base}(Y) = 400$ and there are $w = 4$ words per memory location. Find the addresses of $Y[5,10]$ and $X[3, 20]$ assuming (i) Y is stored in row-major order and (ii) X is stored in column-major order.

Part 2:

Suppose a company keeps a linear array YEAR (1920:1970) such that YEAR [K] contains the number of employees born in year K. Write algorithm for each of the following tasks:

- To print each of the years in which no employee was born.

- b. To find the number NNN of years in which no employee was born.
- c. To find the number $N50$ of employees who will be at least 50 years old at the end of the year. (Assume 1984 is the current year).
- d. To find the number NL of employees who will be at least L years old at the end of the year. (Assume 1984 is the current year).